TRIBHUVAN UNIVERSITY

FACULTY OF HUMANITIES AND SOCIAL SCIENCES

A Project Report

On

“MovieMagic – “The Streaming Aspect for Cinematography”

Submitted to:

Department of Computer Application

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Submitted By:

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ABSTRACT

This project aims to develop an online movie streaming platform that offers users a convenient and reliable way to watch movies, shows, and other video content. The platform will feature a rich library of movies, TV shows, and documentaries, with regular updates to ensure fresh and engaging content. Users will be able to access the platform from various devices for maximum flexibility. Core features include personalized movie recommendations based on user preferences and viewing history, as well as a section highlighting trending content based on the user's locality. Additionally, the platform will support user reviews and ratings, allowing viewers to share their opinions and help others discover popular content. Designed with security and performance in mind, the platform aims to deliver a seamless and satisfying streaming experience.

Table of Contents

LIST OF ABBREVIATIONS

CSS: Cascading Style Sheet

HTML: Hypertext Markup Language

MySql:Database Management System

PHP: PHP Hypertext Preprocessor

SQL: Structured Query Language

LIST OF FIGURES

LIST OF TABLES

Chapter 1:
Introduction

1.1 Introduction

MovieMagic Web-based movie streaming platform Users can stream a large number of movies and television segments directly from the website without requiring cable or satellite service. However, the screen users are greeted with is fairly simple, with clearly defined categories based on genre and other factors, along with a description of the content before watching it.

It includes features such as user review and rating options, among others, to keep users engaged on the platform. MovieMagic provides users with the ability to stream quality video from anywhere at the touch of a fingertip using web browsers on internet-connected devices. The project caters to a wide variety of tastes by providing a well-rounded and modern streaming experience focused around simplicity, discovering new relevant content, and allowing for user input.

1.2 Problem Statement

With the increasing number of movies available online, users often find it hard to decide what to watch. Many streaming platforms lack features that suggest movies based on a user's interests. This leads to wasted time and a less enjoyable experience. Users also miss out on trending content that could match their taste. A basic platform that shows popular and relevant movies can help solve this issue. Such a system would make content discovery easier and more engaging. There is a clear need for a simple movie streaming website that helps users find the right content quickly.

1.3 Objective

The main objective of this project is:

- To create a website for streaming movies that helps users find interesting content and stay engaged by showing popular movies and suggestions based on their interests

1.4 Scope and Limitation

MovieMagic is a web-based streaming platform accessible via desktop, laptop and mobile browsers. It offers a categorized library of movies and TV shows, complete descriptions, and genre filters. Key features include a review and rating system to encourage user interaction. The platform ensures secure login and smooth streaming, built using HTML, CSS, JavaScript, PHP, and MySQL.

The platform is limited to web access and does not support mobile apps or offline viewing. Features like live streaming, or paid subscription models are not included. Content availability is limited due to the absence of official licensing.

1.5 Report Organization

Chapter 1: It includes the introduction section, the problem we attempted to solve, the objectives and scope.

Chapter 2: It includes the background study of fundamental theories, general concepts and literature review of similar projects, theories and results by other researchers.

Chapter 3: It provides an overview of all the requirements along with system analysis and feasibility analysis of the system.

Chapter 4: It includes a detailed description of how the system was designed.

Chapter 5: It includes the tools we used to build the system and how the testing process was done.

Chapter 6: It includes the conclusion of the project and how we are further planning to make the system work sustainably.

Chapter 2:

Background Study and Literature Review

2.1 Background Study

Popular streaming services like Netflix offer a wide range of content and features, but they often come with high subscription costs. Free alternatives usually lack useful features like smart suggestions or user interaction. These limitations inspired the idea of MovieMagic — a simple and engaging movie streaming platform. The aim was to create a site that focuses on content suggestions based on genres, along with user reviews and ratings. The platform also keeps a history of watched content to help users keep track of what they've seen. MovieMagic is designed to make discovering and enjoying movies and TV shows easier and more fun for users.

2.2 Literature Review

The evolution of video streaming platforms has significantly transformed media consumption patterns over the past decade. Studies and articles related to platforms like Netflix highlight the importance of user personalization [7] [8], with machine learning algorithms playing a crucial role in recommending content based on watch history and preferences. However, such systems often require complex infrastructure and large datasets, making them less feasible for smaller or independent platforms.

Research into free streaming sites reveals a focus on wide content availability but also points out drawbacks [9] [10] [2] such as lack of user engagement features, poor content categorization, and intrusive advertising. These issues affect user retention and overall satisfaction. In contrast, academic papers on streaming emphasize that clean design, localized content suggestions, and minimal friction during navigation significantly improve viewer engagement.

Open-source APIs such as YTS, TMDB, and OMDb have also been explored in existing literature for their usefulness [6] [5] [4] in educational or experimental projects. These sources provide structured data on films, including metadata, trailers, posters, and more, enabling developers to implement dynamic content integration without hosting large media files.

From this review, it is evident that a successful streaming platform must balance accessibility, personalization, and clean UI. Drawing from the strengths and limitations of existing platforms and literature, MovieMagic aims to deliver a lightweight, informative, and user-driven streaming experience focused on core features like user feedback systems.

Chapter 3:

Requirement Analysis and Feasibility Study

3.1 System Analysis

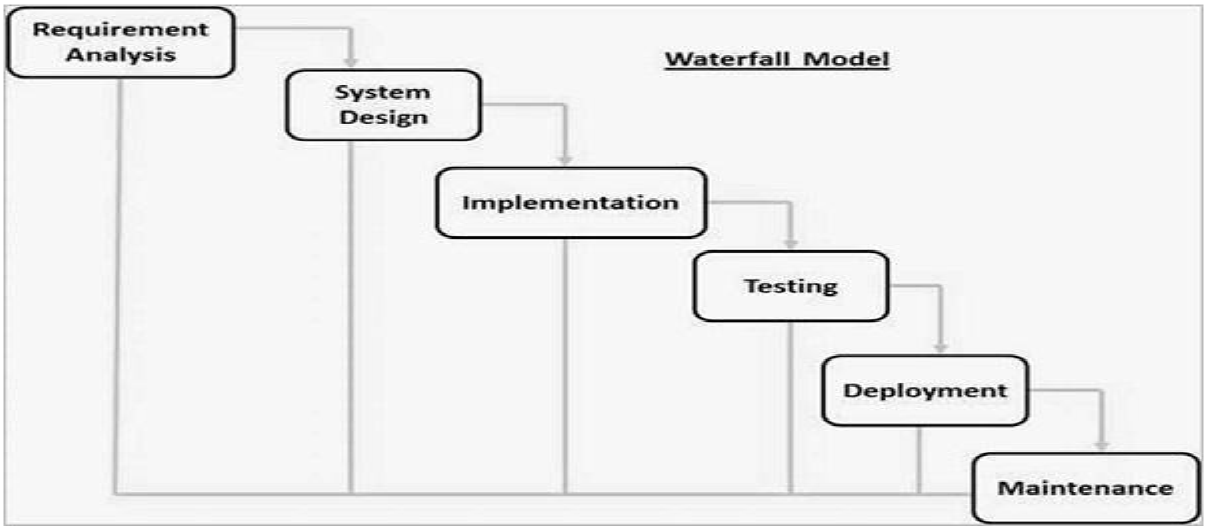

Figure 1: Waterfall Model

The development of MovieMagic follows the Waterfall Model, a linear and sequential approach [11] that ensures each phase is completed before moving to the next. This model is ideal for projects with well-defined requirements and allows for thorough planning, design, and implementation with minimal risk of major changes during development.

3.1.1 Requirement Analysis

The goal of the requirement analysis process for this project was to determine the essential functions and features of the system as well as any restrictions or limits that would need to be taken into account when it was being developed.

3.1.1.1 Functional Requirement

The movie streaming platform must include a collection of movies and TV shows with genre-wise organization and efficient video playback management. The system can only be accessed by the admin. Movies and information can be added, removed, or edited by the admin. Movies may be streamed and browsed online.

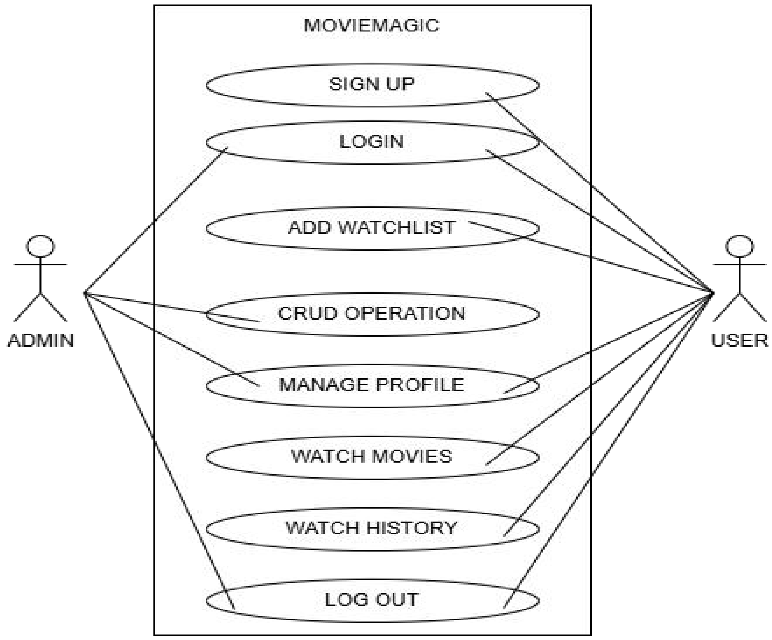

Figure 2: Use case diagram of MovieMagic

3.1.1.2 Non-functional Requirement

The non-functional requirements for the movie streaming platform focus on ensuring reliability, performance, and security. The platform should provide high availability and fault tolerance to minimize downtime and ensure continuous access to content. Performance metrics include fast load times and seamless video playback with minimal buffering, supported by a scalable infrastructure to handle varying levels of user traffic and large volumes of video data. The system must also be maintainable and extendable, allowing for future upgrades and the integration of new features without significant disruption.

3.1.2 Feasibility Study

The feasibility study assesses the system's functionality under specific constraints, evaluating how viable it is to develop the system under these conditions. The key feasibility aspects considered include operational, economic, and technical feasibility.

3.1.2.1 Technical Feasibility

MovieMagic is technically feasible with the available features like video playback, genre-based browsing, user reviews, and history tracking. These can be developed without advanced tools or external dependencies. The system design is simple, focusing on admin-controlled content and user interaction. All required elements can be implemented using existing knowledge and resources. Therefore, the project can be successfully completed within the current capabilities.

3.1.2.2 Operational Feasibility

The MovieMagic platform is operationally feasible as it requires minimal resources to run and maintain. It can be implemented using available systems, such as a basic server or local hosting environment. Since content is managed by the admin and users mainly stream and browse, the overall operation remains simple. The system's core functions—like video playback, genre filtering, reviews, and history—can be handled smoothly without the need for high-end infrastructure. Given the current resource availability, the project can operate efficiently and meet its intended goals.

3.1.2.3 Economic Feasibility

MovieMagic is economically feasible as it involves low development and operational costs. Since it is a project-based platform, it does not require expensive infrastructure or licensing. The system can run using freely available resources, and there are no ongoing subscription or hosting fees in the current setup. While it is not designed for commercial revenue at this stage, it demonstrates potential for future growth if expanded. Considering the minimal financial input and available resources, the project is financially viable in its current form.

3.1.3 Data Modeling

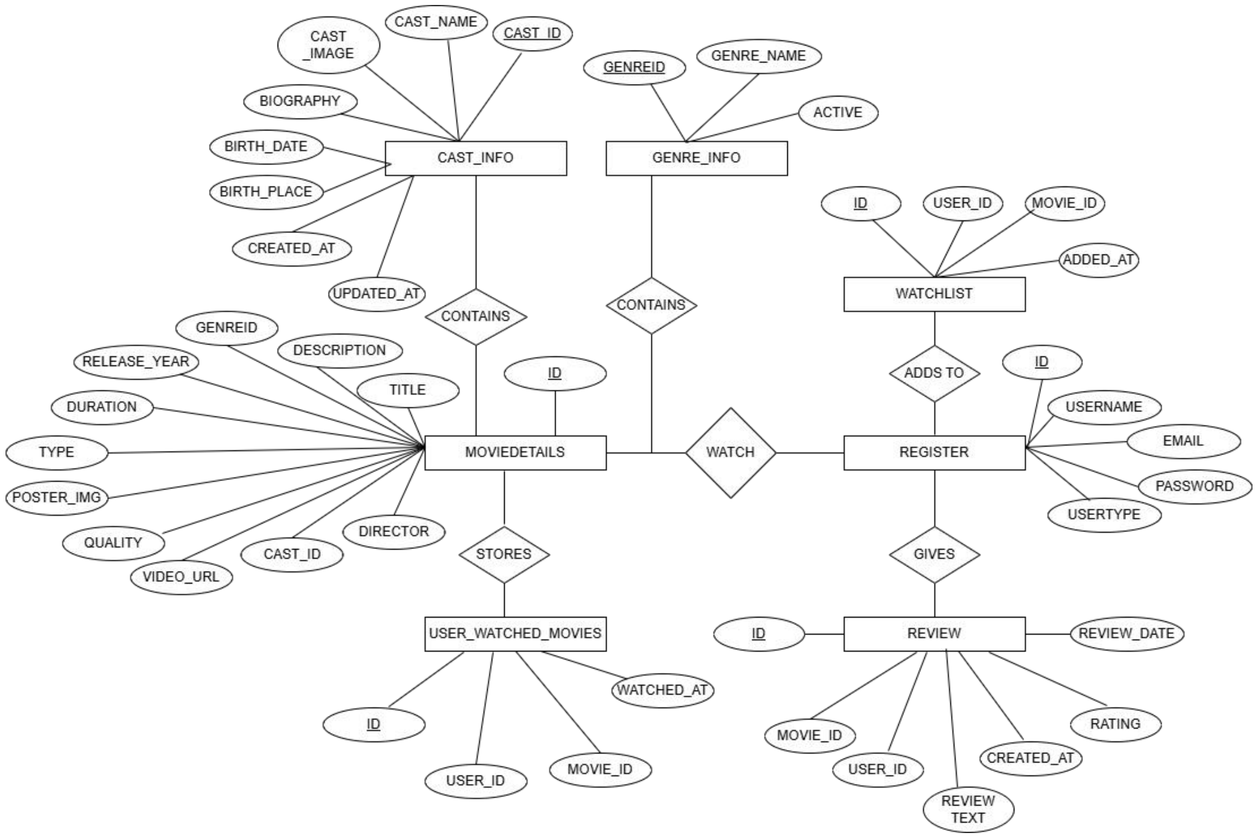

Figure 3: ER-Digram of MovieMagic

The ER diagram represents a movie database with the following entities:

- **Register:** Represents a user or an admin based on the user type.
- **MovieDetails:** Represents a movie with information like title, description, genre, etc.
- **Cast Info:** Represents details of an Actor.
- **Watchlist:** Represents a list of movies a user wants to watch.
- **Reviews:** Represents user reviews of a movie.
- **User Watched Movies:** Represents a list of movies a user have watched.
- **Genre_info:** Provides information about the genre.

3.1.4 Process Modeling

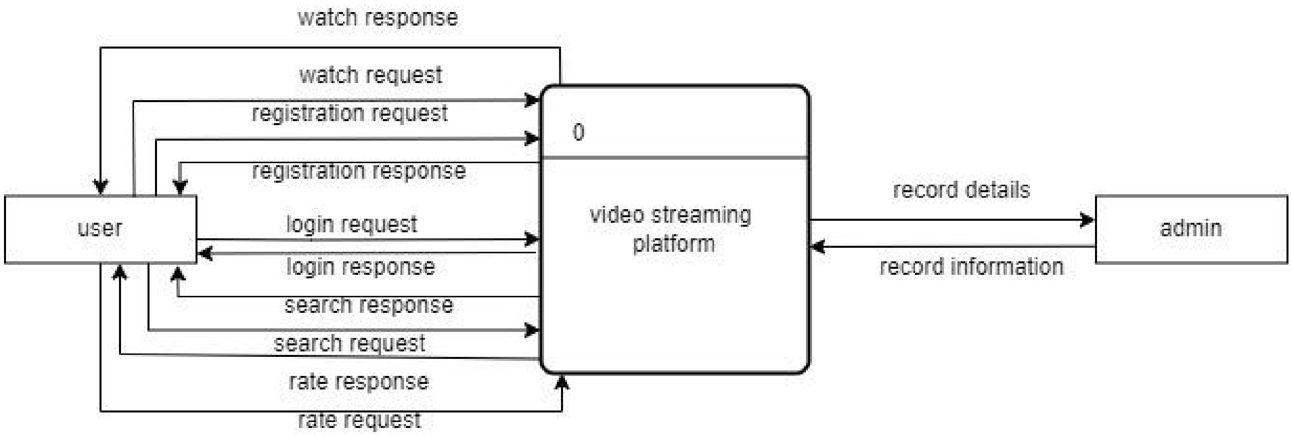

Figure 4: DFD level-0 of MovieMagic

The context diagram provides a top-level view of how the system interacts with its environment and the data it processes. It does not go into the details of internal processes; instead, it focuses on the high-level relationships between the system and external entities. It interacts with the user as:

- From Users to System: login credentials, registration information, search queries, streaming requests.

From System to Users: streamed content, user account information.

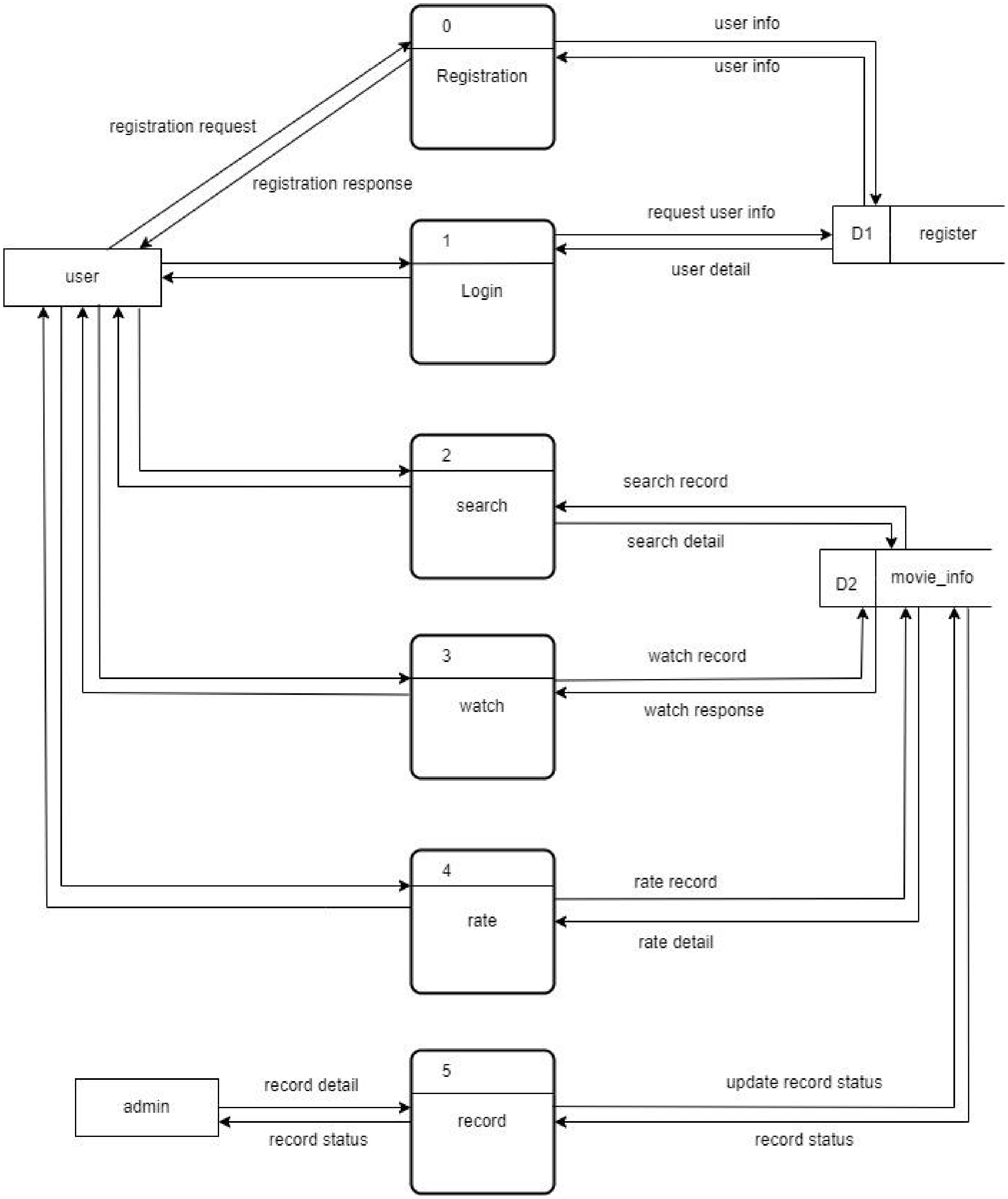

Figure 5: DFD level 1 of MovieMagic

This Level 1 DFD provides an overview of how different processes interact with data stores and external entities. It outlines the flow of data within the system and the basic functionality of the various processes.

1. Process: User Management

- Register User: Collects user information (username, password, email) and saves it in the user Database.
- Login User: Authenticates user credentials from the User Database and grants access.

2. Process: Streaming Service

- Stream Movie: Streams requested movies to users.

3. Data Stores

- User Database: Stores user registration and login data.
- Movie Database: Stores movie information including titles, descriptions, genres, and streaming links.

External Entities:

- Users:
  - Provide registration information, login credentials, and movie requests.
  - Receive streamed movies, movie information.
- Admin:
  - Provides new movie data, updated movie data, and delete requests.
  - Receives success or error messages for movie creation, updates, and deletions.

3.1.5 Flow Chart for Admin:

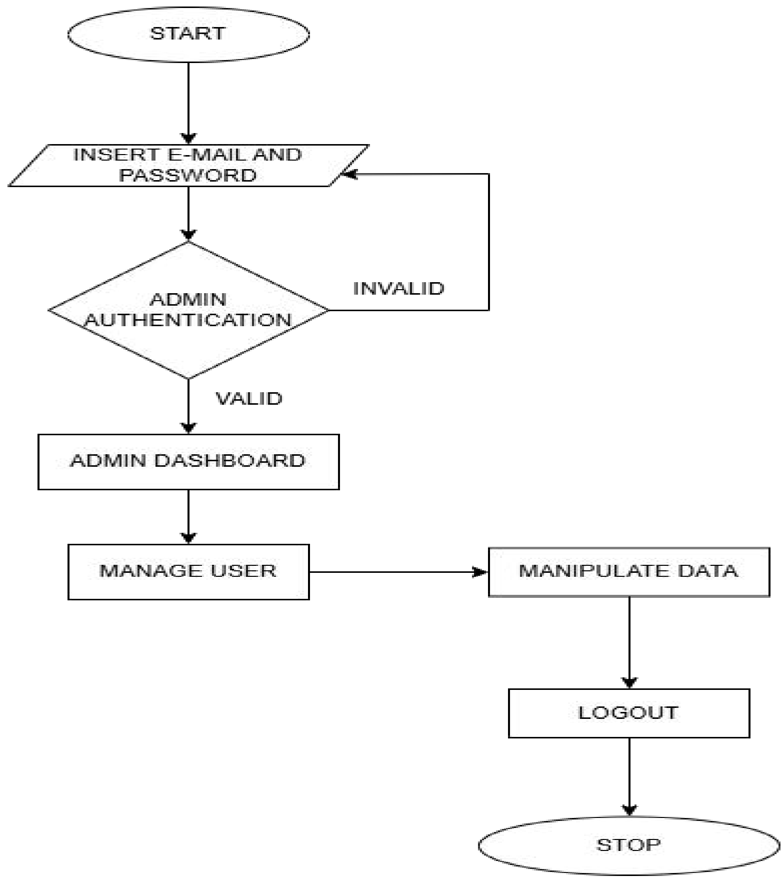

Figure 6: Flow-chart of Admin

In the above figure 3.6, flowchart diagram of admin where admin can manage and handles the User Information and Movie/TV-Shows details.

3.1.5 Flow Chart for User:

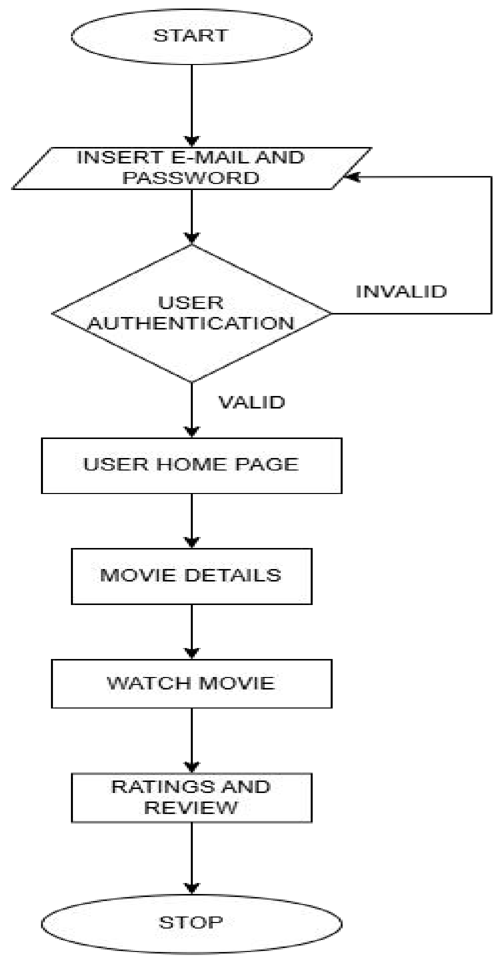

Figure 7: Flowchart of User

In the above figure 3.7, The user flowchart diagram above allows users to watch movies and TV series based on their preferences, complete with descriptions, trailers, and other details about the actors, directors, and writers.With a personalized recommendation system, one can also view the ratings and reviews of the films left by other users.

Chapter 4:

System Design

4.1 Design

The movie streaming platform has a user-friendly style that makes it simple to navigate on desktop. Its easy-to-use movie and TV program classification improves user accessibility. Personalized suggestions and convenient access to stored material are made possible by user profiles. A secure and seamless user experience is ensured by consistent branding and strong security measures that safeguard user data and transactions.

4.1.1 Architectural Design

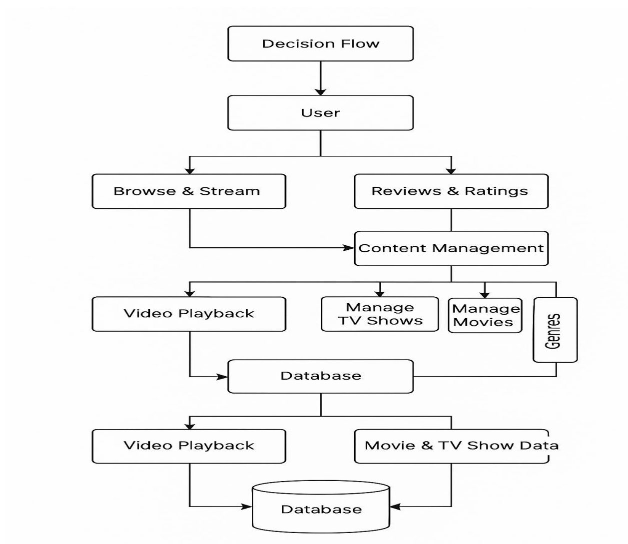

Figure 8: Architectural Design of MovieMagic

An architectural diagram for movie streaming website provides an overview of the system's structure, including its major components, their interactions, and how data flows between them. This kind of diagram helps you understand how the different parts of the system work together to achieve the desired functionality.

Key components:

1. Top-Level (Decision Flow → User)

- Decision Flow represents how the system determines what actions can be taken.

- It leads to the User, who interacts with the system—either browsing, streaming, or leaving reviews.

2. User Activities:

- **Browse & Stream**: Users can explore available content and stream movies or TV shows.
- **Reviews & Ratings**: Users can leave feedback on the content they watch.
- Both these activities feed into **Content Management**.

3. Admin/Content Management Section
- Content Management: Handled by the Admin (though not explicitly shown, it's implied here). This is where the admin manages content updates.
  - Manage TV Shows
  - Manage Movies
  - These are organized by Genres (like action, drama, etc.).
4. Video Playback & Database
- The Video Playback module ensures content is streamed properly.
- All content and activity connect to a Database, which stores:
  - Movie & TV Show Data
  - User Reviews, Watch History, etc.
- The Database supports playback, content browsing, and management.
5. Bottom Layer
- The system loops back to support Video Playback and content delivery, showing a two-way flow of data between the interface and the database.

4.1.2Algorithm Details

Collaborative Filtering Algorithm

In the future, MovieMagic plans to implement a recommendation system using the Collaborative Filtering algorithm to enhance user engagement and content discovery. This algorithm is chosen because it does not rely on the content details of movies but instead uses user behavior such as ratings and reviews, which are already part of the platform. Collaborative Filtering works by identifying patterns among users with similar preferences. For example, if two users have watched and liked the same set of movies, the system can recommend new movies based on what one user has seen and the other hasn't. This approach helps deliver personalized suggestions without manually tagging or analyzing content. It plays a crucial role in making the platform smarter and more user-focused, helping users find relevant content quickly and keeping them engaged over time.

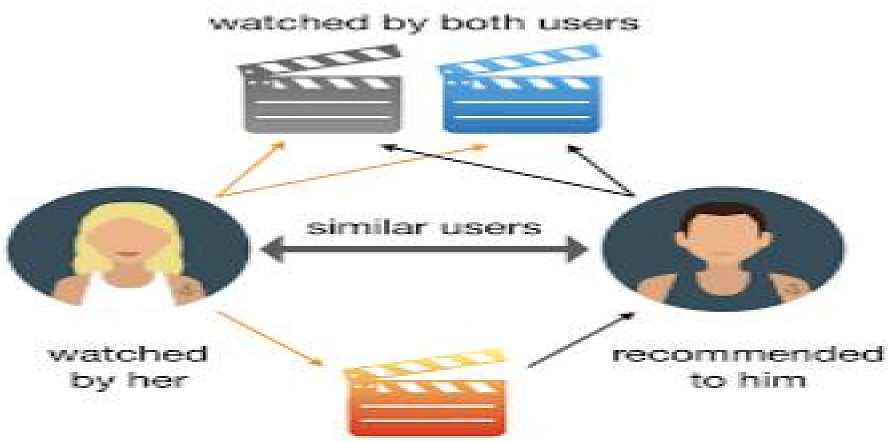

Table 1:cast_info

cast_id	int(11)AUTO_INCREMENT PRIMARY KEY
cast_name	varchar(100)
cast_image	varchar(255)
biography	text
birth_date	date
birth_place	varchar(100)
created_at	timestamp
updated_at	timestamp

Table 2:genre_info

Genreid	int(11)AUTO_INCREMENT PRIMARY KEY
genre_name	varchar(50)
Active	int(4)

Table 3:moviedetails

id	int(11) AUTO_INCREMENT PRIMARY KEY
title	varchar(191)
description	Longtext
genreid	int(11)
release_year	int(11)
duration	varchar(30)
type	varchar(30)
poster_img	varchar(191)
quality	Mediumtext
video_url	varchar(255)
cast_id	int(11) Foreign Key
director	varchar(100)

Table 4:register

id	int(11)AUTO_INCREMENT PRIMARY KEY
username	varchar(50)
email	varchar(150)
password	varchar(50)
usertype	varchar(20)

Table 5:reviews

id	int(11) AUTO_INCREMENT PRIMARY KEY
video_id	int(11)
user_id	int(11)
review_text	Text
rating	tinyint(4)
created_at	Timestamp

review_date	timestamp

Table 6:user_watched_movies

id	int(11) AUTO_INCREMENT PRIMARY KEY
user_id	int(11) Foreign Key
movie_id	int(11) Foreign Key
watched_at	timestamp

Table 7:watchlist

id	int(11) AUTO_INCREMENT PRIMARY KEY
user_id	int(11) Foreign Key
movie_id	int(11) Foreign Key
added_at	timestamp

Here's a brief explanation of some of the terms in the context:

title, description, id, name, usertype, register, password,username,email, genreid, genre_name, genre_info,active,moviedetails, video_url,quality, poster_img, release_year, type, duration

These terms are common in movie or TV show databases and refer to various attributes associated with a movie or TV show.

- title: The name of the movie or TV show.
- description: A brief summary of the movie or TV show's plot.
- id: A unique identifier for the movie or TV show.
- usertype: The type of user account, such as a admin or user.
- register: The action of registering for a user account.
- password: The secret code used to authenticate a user account.
- username: A unique name used to identify a user account.
- email: The user's email address.
- genreid: A unique identifier for the genre of the movie or TV show.
- genre_name: The name of the genre of the movie or TV show.

- genre_info: Additional information about the genre of the movie or TV show.
- active: A flag indicating whether the movie or TV show is currently available.
- moviedetails: Detailed information about the movie or TV show.
- video_url: The web address of the movie or TV show's page.
- quality: The video quality of the movie or TV show.
- poster_img: The image used to represent the movie or TV show.
- release_year: The year the movie or TV show was released.
- type: The type of media, such as a movie or TV show.
- duration: The length of the movie or TV show.

Chapter 5:
Implementation and Testing

5.1 Implementation

The waterfall model is applied to develop a movie streaming platform through a systematic, sequential approach. It begins with gathering comprehensive requirements from stakeholders and users, detailing functionalities like content categorization, playback quality, and user interface design. With these requirements in hand, the platform's architecture is meticulously designed, encompassing frontend layout, backend infrastructure for data storage, and scalability considerations for future growth. Development follows, where software engineers write code and integrate necessary components, ensuring each phase is completed before moving to the next. Rigorous testing is then conducted to verify functionality and performance, ensuring seamless user experience and reliability in streaming services. Once tested and approved, the platform is deployed, and ongoing maintenance ensures it meets user expectations while adhering to security standards and evolving industry trends.

5.1.1 Tools Used

For our movie streaming platform project, a combination of essential tools and technologies is employed across various development phases:

Frontend:

- **HTML:** Utilized to structure and define the content and layout of web pages, including movie listings, user profiles, and streaming interfaces.
- **CSS:** Used for styling and enhancing the visual presentation of the platform, ensuring a cohesive and user-friendly design.
- **JavaScript (JS):** Implements interactive features and dynamic content updates, enhancing user engagement and functionality across different devices.

Backend:

- **PHP:** Powers server-side operations and dynamic content generation, handling user authentication, session management, and database interactions.

Database:

- **MySQL:** Serves as the database management system to store and manage movie metadata, user data, and transactional information efficiently.

Development Environment:

- **Visual Studio Code (VSCode):** Chosen as the primary integrated development environment (IDE) for coding frontend and backend components. VSCode provides robust editing capabilities, debugging tools, and extensions for seamless workflow management and collaboration.

    - By leveraging these tools effectively, MovieMagic aims to deliver a responsive, secure, and feature-rich experience for users, ensuring efficient data management and seamless interaction across different devices and platforms.

Chapter 6:

Conclusion and Future Improvements

6.1 Conclusion

MovieMagic aims to provide a smooth and engaging movie streaming experience by focusing on essential features like video playback, genre-based content, user reviews, and viewing history. While the current version includes the core functionalities, there is scope for future enhancements like personalized recommendations. Overall, the platform offers a solid foundation for further development and a better user experience.

6.2 Future Recommendations

In addition to personalized recommendations and locality-based trends, MovieMagic can be enhanced with several intelligent features using various algorithms. A mood-based recommendation system can be implemented using sentiment analysis to suggest movies based on the user's current mood or emotional state. Watch-time prediction algorithms can help estimate the best time slots for users to watch content, improving engagement. A dynamic content ranking system can also be introduced, which adjusts the order of displayed content based on user interactions and popularity metrics using ranking algorithms. Furthermore, implementing automated content tagging through machine learning can simplify content management by generating genre tags, cast details, and themes. These features would make the platform smarter, more interactive, and adaptable to user preferences over time.

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